

Modeling Wildfire Spread in California with Cellular Automata

Team 4: Gabriel Appiah, Katherine Losada, Kshitij Sawant, and Thanawit
Suwannikom

GitHub Repo: <https://github.gatech.edu/tsuwannikom3/cse6730-project>

1. Abstract

This project develops a GIS-driven cellular automata framework to simulate wildfire spread across a California case study using empirical land-use and fuel data. We discretize the study area into grid cells classified as one of four states: no fuel, unburned fuel, burning, and burned completely. State transitions occur in discrete time steps based on probabilistic functions informed by (1) land-cover classes derived from California land-use data, (2) vegetation type and density from satellite-derived NDVI rasters, (3) dynamic wind fields, and (4) terrain elevation. Ignition probabilities follow a logistic-regression model calibrated with historical fire occurrences and human-activity indicators. By replacing synthetic inputs with real-world datasets, the simulation more accurately captures the complex dynamics of environmental drivers and wildfire behavior. This framework provides emergency responders and environmental agencies with a robust tool for understanding and forecasting fire behavior, supporting informed decision-making during wildfire events.

2. Project Description

Wildfires have been more frequent and destructive in recent years. The wildfire in Los Angeles, which took place throughout the month of January this year is considered to have occurred outside the usual wildfire season, but the most destructive in US history, costing over 250 billion [1]. These unparalleled events are driven by the changes in climate, land cover, and sociodemographic conditions [2]. The frequency and intensity of wildfires are expected to increase in the future, as such the ability to properly model and simulate these events is vital. Fone [3] was the first person to use mathematical models to describe the spread of fire. He modeled the spread of fire as a series of successive ignitions influenced by the ignition time and the distance between particles making the problem easier to analysis. Since then, several scholars have simulated the spread of fire, mostly representing it as partial differential equations (PDE), discretized in the form of finite differences or finite elements [4].

The primary goal of our project is to develop a simulation model that accurately depicts the spread of wildfires using a cellular automata (CA) framework. The simulation aims to serve as a predictive tool that deepens our understanding of wildfire behavior, especially in relation to the surrounding environmental conditions. We demonstrate our model by simulating wildfire propagation in California using Geographic Information Science (GIS) datasets. Through this experiment, we explore how different factors (terrain slope, wind speed, wind direction, land use, vegetation type, and vegetation density) influence the propagation of wildfire and how varying the duration for the state transition of the grid cells impact the spread of wildfire. This project intends to answer the following questions:

- a. How do environmental factors such as slope, wind speed, and land cover influence the rate and direction of wildfire spread in California?
- b. Which regions in the study area exhibited the highest frequency of burning across multiple simulation runs, and what environmental factors explain this pattern?

3. Literature Review

3.1 Factors Defining and Affecting Wildfires

While defining wildfires, the core characteristics include the temperature of the fires themselves, the speed at which the wildfire is spreading, as well as the area affected. The type of tree canopy being affected plays a role in defining the type of wildfires that are being produced. [5] Fires consuming organic matter beneath the ground such as roots, fallen leaves, branches lead to Ground fires, the fires consuming similar matter on the forest floor can cause Surface fires, and when the fire is consuming trees and vegetation well above the forest floor, high-intensity fires called Crown fires occur. The wind direction relative to the wildfire determines whether it is a Heading fire (moves in the direction of the wind) or a backing fire (moves opposite to the direction of the wind) [6].

The presence of higher wind supply itself is an enabler to spread fire. Other than wind speed and direction, localities with higher temperatures and low humidity lead to drier conditions which increases the probability of a wildfire originating in the area. The type of fuel contributes to how much combustible material is available to burn, as dead and dry vegetation burns more easily than live, moist fauna. [7] Weather conditions and precipitation levels contribute to factoring as regions with heavy rainfall lead to higher vegetation growth and hence higher fuel available, and areas with prolonged droughts drying out available vegetation and create a fire risk. In relation to topology, fires burn faster when they are either uphill or are funneled winds through valleys and canyons. Presence of water bodies reduces the likelihood of ground fires as well as relate to higher precipitation. Human factors include campfires, fireworks, agricultural burning or electrical sparks caused by neighboring power lines can spark a wildfire. Hence the expansion of urban areas into wildland increases the likelihood of human caused fires [8].

3.2 Methods Used in Simulating Wildfires and Optimizing Emergency Response

Beyki et al. [9] reviewed different wildfire evacuation simulation methods, classifying fire spread models into physical, empirical, and numerical approaches. Physical models, like Rothermel's [10] widely recognized equation, use physics-based principles to estimate fire behavior. Empirical models rely on statistical patterns from past fires. Cellular automata (CA), a semi-empirical approach, simulates fire spread through grid-based rules that approximate real-world dynamics. The authors also examined computational fluid dynamics (CFD) models, which offer highly detailed fire simulations but require significant computing power. Evacuation models include agent-based and traffic simulations, as well as GIS approaches like evacuation trigger buffers. Integrated platforms, such as WUI-NITY [11], combine fire spread, pedestrian, and traffic models for a more comprehensive analysis. While the authors did not point to a single best method, they highlighted the value of hybrid approaches that integrate multiple models to better capture fire behavior, human movement, and evacuation logistics.

Burge et al. [12] introduce a deep learning model using Recurrent Convolutional Neural Networks (CNNs) to predict time-resolved wildfire spread behavior. The model makes autoregressive predictions every 15 minutes, capturing the evolving nature of wildfires and incorporating

temporal dependencies to improve predictions over time. It is trained on real-world wildfire datasets from California and simulates fire spread while ensuring computational speed, which is crucial for real-time emergency response. The model also integrates various environmental factors—wind, temperature, and vegetation—into its predictions of wildfire spread. Compared to traditional wildfire simulation methods (such as physical or empirical models), the deep learning model demonstrates potential to improve wildfire management by providing more accurate, real-time predictions.

Cellular Automata (CA) models have been widely used in wildfire simulation due to their ability to represent fire spread dynamics through simple, rule-based interactions on a grid. Ghisu et al. [13] developed an optimized CA algorithm that improves computational efficiency while maintaining accuracy in fire propagation modeling. Trunfio et al. [14] introduced a novel CA-based wildfire simulation algorithm that enhances fire spread predictions by incorporating local environmental conditions and probabilistic transition rules. Similarly, Velásquez et al. [15] integrated CA with GIS tools to create a wildfire propagation simulation framework, demonstrating its applicability in real-world spatial analysis. These studies highlight CA's flexibility in modeling complex fire behavior while balancing computational demands, making it a valuable tool for wildfire prediction and management.

Jiang et al. [16] introduced an Irregular Graph Network (IGN) that adapts node density based on terrain complexity, using a deep learning model (WFDNN) to predict fire spread duration between nodes. This approach demonstrated superior spatiotemporal tracking of fire progression compared to traditional grid-based models like FARSITE. Similarly, Rösch et al. [17] developed a spatiotemporal graph neural networks (GNN) to model fire spread across European landscapes, showing promising generalizability across different regions despite limitations in available fire records.

Beyond prediction, other studies have explored graph-based decision-making and simulation for wildfire management. Sá et al. [18] integrated graph connectivity analysis with fire spread models to identify high-risk fuel patches that facilitate fire propagation, offering insights for targeted fuel management. Meanwhile, Aveiro et al. [19] used graph-theoretic optimization to model fire suppression as an edge-removal problem, identifying optimal locations for intervention to block fire spread.

4. Conceptual Model

We developed a CA model that uses mathematical foundations for modeling the propagation of wildfire spread across a landscape like the one created by Velásquez et al. [15]. As part of this CA model, we structure the landscape into a grid, with each cell representing a small section. These grids can exist in four states: no fuel (empty), contains fuel but not ignited, burning, and burned completely. The fire spreads across the grid following a set of transition rules for each discretized time step. Based on environmental conditions and probability calculations, these rules determine whether neighboring cells catch fire. The transition rules are applied based on the Moore Neighborhood (a central cell with its eight surrounding neighbors). Fire spread occurs when a

burning cell transmits fire to one or more of its adjacent vegetated neighbors, conditional on the presence of flammable fuel and influenced by multiple factors. The following are the transition rules for a given cell (i, j) with each time step t :

-Rule 1: If the state of $(i, j, t) = 1$, then state of $(i, j, t+1) = 1$. This indicates a cell with no fuel will always remain the same as these cells cannot burn.

-Rule 2: If the state of $(i, j, t) = 3$, then state of $(i, j, t+1) = 4$. If the current state of the cell is 3 which indicates burning, then in the next time step the cell will transition to a burned state.

-Rule 3: If the state of $(i, j, t) = 4$, then state of $(i, j, t+1) = 4$. A cell which is in a burned state will remain in that state for the subsequent time steps

-Rule 4: If the state of $(i, j, t) = 3$, then state of $(i \pm 1, j \pm 1, t+1) = 3$, with a certain probability of fire.

The ignition of fire within the grid is based on an ignition probability model. This model estimates the ignition probability at each cell using a logistic function of predictor variables. The spread of the fire is governed by a probability formula developed by Alexandridis, et al. [20], which depends on factors such as wind speed, wind direction, slope, land use/ vegetation type, and vegetation density. The direction and spread of wind influence how quickly and in which direction the fire moves. Also, terrain elevation affects the spread of fire: fire spreads quickly uphill due to the rising heat aiding combustion, and more slowly downhill. Finally, vegetation density influences the propagation of wildfires, with denser forests providing the fuel for the fire's spread. The conceptual model captures the essential dynamics of wildfire behavior in a computationally tractable yet spatially explicit form, allowing for scenario testing, and sensitivity analysis for a fire-prone area.

5. Simulation Model

Our simulation model is a comprehensive framework for predicting wildfire spread across a discretized landscape grid using cellular automata. The system is primarily built using Python, leveraging libraries such as NumPy for matrix-based fire propagation modeling, Matplotlib for visualization, and Requests for API integration with external data sources like Google Maps API for geographical data and OpenWeather API for real-time weather conditions. The simulation can be executed in Jupyter Notebook or as a standalone Python application, making it highly flexible for research and operational use. Additionally, the tool can be extended with GIS software such as QGIS for spatial analysis and Google Earth Engine for real-time terrain and vegetation monitoring.

The simulation model integrates components — ignition dynamics, wildfire spread probability equation, vegetation type, vegetation density, wind speed/direction, and terrain slope — to simulate the progression of wildfires under various environmental conditions. The following

subsections describe each component independently and the verification procedures used to ensure the implementation aligns with the conceptual model.

5.1 Ignition

We implemented the ignition model to simulate a spatial ignition probability map inspired by the logistic regression model developed by Parisien et al. [22], which integrates multiple environmental and anthropogenic factors to estimate the likelihood of wildfire ignitions across a landscape. The model follows a logistic-style formulation (see formula below) where the ignition probability at each grid cell is computed as a nonlinear combination of predictor variables such as vegetation density, slope, lightning density, proximity to roads, land use type, historical fire occurrences, temperature, and dryness index. Each of these variables contributes to the spatial heterogeneity of ignition potential, and their respective influences are weighted according to their relevance as suggested in the literature.

$$P(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}}$$

Where

$P(x)$ is the probability of ignition occurring at location x

$x_1, x_2, \dots, x_n$ is the predictor variables

We first prototyped a model without actual geospatial datasets, where each input parameter is generated using random distributions that mimic real-world patterns. We next applied inputs, including Normalized Difference Vegetation Index (NDVI), road proximity, Normalized Difference Drought Index (NDDI), surface temperature, slope, land use, lightning density (synthetic), and wildfire history (synthetic), to estimate the ignition probability for the study area as part of implementing the simulation model with an actual GIS dataset. Since we could not get appropriate data for lightning density and historical wildfire, we used simple methods from the literature to represent them. For example, the lightning density in California seems constant overall, according to Vaisala Xweather [21].

5.2 Wildfire Spread Probability Equation

To simulate how fire spreads across the landscape, we use a probabilistic equation that accounts for key environmental variables influencing ignition likelihood. The fire spread probability for each cell in the cellular automata model is calculated using the following equation, allowing the model to reflect the spatial variability and randomness inherent in wildfire behavior.

$$P_{burn} = P_0(1 + P_{veg})(1 + P_{den}) \cdot P_w \cdot P_{ele}$$

Fire propagates from a burning cell to its neighboring cells with a probability P_{burn} which depends on environmental factors.

$$P_{burn} = P_0(1 + P_{veg})(1 + P_{den})P_wP_{ele}$$

Where:

$P_0 = 0.58$ (empirical value)

P_{veg} is vegetation type factor

P_{den} is vegetation density factor

P_w is wind influenced factor

P_{ele} is terrain elevation factor

$$P_w = e^{c_1 V} e^{c_2 V (\cos \theta - 1)}$$

Where:

$c_1 = 0.045$, $c_2 = 0.131$ (empirical values)

V is wind speed (m/s)

θ is the angle between wind direction and fire spread direction

$$P_{ele} = e^{a\theta_s}$$

Where:

$a = 0.078$ (empirical value)

$\theta_s = \arctan\left(\frac{E_1 - E_2}{l}\right)$ for adjacent cell

$\theta_s = \arctan\left(\frac{E_1 - E_2}{l\sqrt{2}}\right)$ for diagonal cell

Where E_1 , E_2 are elevations of neighboring and center cells, and l is the distance between two cells.

5.3 Vegetation Type

We determined vegetation type categories using NDVI thresholds and incorporated literature-based burn rate modifiers from Alexandridis, et al. [20]. NDVI values were normalized from a GeoTIFF file and then classified into four categories: NDVI < 0.2 is considered “no vegetation,” values between 0.2 and 0.4 are assigned to shrub/grassland, 0.4 to 0.6 represent agriculture, and

values above 0.6 are classified as trees/forest. We acknowledge that NDVI thresholds for vegetation types vary across sources, so some misclassification may occur—especially for agriculture, whose NDVI scores can overlap with those of trees. Additionally, the thresholds were adjusted to ensure continuous classification with no unclassified values.

The vegetation type Pveg burn rate modifiers were directly taken from Alexandridis, et al [20]. (2008), with agriculture assigned -0.3, shrub/grassland set to 0, and forests assigned 0.4. These modifiers adjust the base fire spread probability in the simulation model. We tested the simulation by running the base fire spread model with these additional Pveg and vegetation type values, first using a per-pixel classification approach and then aggregating pixels into larger grid cells with renormalized NDVI values. This allowed us to compare the high-resolution classification with an aggregated version, ensuring the robustness of our methodology in capturing fuel variability and its effect on fire propagation.

5.4 Vegetation Density

Vegetation density plays a critical role in modeling wildfire behavior, as it directly impacts the amount of available fuel and influences the intensity and rate at which a fire spreads. In this project, vegetation density is modeled as a dynamic parameter within the cellular automata (CA) framework to reflect real-world environmental conditions more accurately. Initially, the team implemented a simplified model by randomly assigning vegetation density levels to each cell in the simulation grid. These levels were encoded using three discrete categories: low (-0.4), medium (0.0), and high (0.3), based on the methodology described by Alexandridis, et al [20]. These values were incorporated into the fire spread probability equation, modifying the likelihood of ignition for each cell and enabling early testing of the impact of density on fire propagation patterns.

The project transitioned from randomized density values to a data-driven approach using NDVI derived from Landsat 09 satellite data product to improve realism. NDVI data provides a quantitative measure of greenness, which correlates closely with vegetation health and density. The team utilized the rasterio library to load the NDVI GeoTIFF file and applied preprocessing steps to normalize and clean the data. A classification function was then used to segment NDVI values into low, medium, and high-density categories, based on either absolute NDVI thresholds or dynamic percentile-based thresholds for better adaptability to specific terrain characteristics. The resulting vegetation density map was resized using `skimage.transform` to align with the resolution of the simulation grid, ensuring each cell's density was informed by corresponding real-world satellite data.

Once integrated, the NDVI-derived vegetation density was used to influence the fire spread probability for each cell in the simulation. Specifically, the density factor (Pden) modified the base ignition probability as part of the core spread equation. This allowed fire to spread more aggressively through areas of dense vegetation while reducing the spread rate in sparse or barren

regions. Visualizations of the density map and corresponding fire simulation results demonstrated more varied and ecologically plausible spread behavior than the randomized model. This step enhanced the simulation's realism and enabled future validation efforts, as areas of known high vegetation can be cross-checked against observed fire activity. Overall, the integration of NDVI-based vegetation density significantly improved the environmental fidelity of the CA model, positioning it as a more robust tool for wildfire risk assessment and emergency planning.

5.5 Wind Speed/Direction

We used dynamic wind speed and direction using data from the Open-Meteo API. This free and reliable weather API provides hourly wind speed (in meters per second) and wind direction (in degrees), based on the meteorological convention where 0° represents wind coming from the north. The wind direction is crucial in influencing fire spread directionality, as fire is more likely to propagate in the direction the wind is blowing toward.

In the simulation model, we updated the fire spread probability function to dynamically adjust based on the angular alignment between the wind direction and the direction of fire propagation between cells. We also implemented a function to interpolate the hourly wind data to finer temporal resolutions (e.g., 5-minute intervals), allowing for smoother and more realistic variation over time. This integration allows the simulation to more accurately reflect the impact of changing wind conditions on the fire's behavior and spread.

5.6 Terrain Slope

We obtained elevation data for each grid cell via the Google Elevation API, constructing a matrix of elevation values over the study area. For each cell, we computed the partial derivatives in the X and Y directions by taking differences between neighboring elevation points. Next, we calculated the slope angle using the arctangent of that gradient magnitude and finally converted the result from radians into degrees. These per-cell slope values feed directly into our terrain elevation factor in the fire-spread probability equation, so that steeper uphill slopes increase spread probability and downhill slopes decrease it.

5.7 Verification

The implementation of each module for simulation were verified with unit tests as described below:

1. **Slope Calculation:** Input a synthetic elevation grid (linear ramp : $z(i, j) = i$) and verify the output slope matches the analytical value (e.g. 45° for a unit gradient). Test edge cases of flat terrain where slope should be zero.
2. **Vegetation Type:** Input a synthetic NDVI array (e.g. [0.1, 0.3, 0.5, 0.7]) and confirm classifications (no vegetation, shrub/grassland, agriculture, tree/forest) and modifiers (-0.3, 0, -0.3, 0.4) are correct.

3. **Vegetation Density:** Input a synthetic raster image of known NDVI value and test the classification into low, medium and high categories.
4. **Wind Speed and Direction:** Mock API data (e.g. wind speed = 8 m/s, direction = 315°) and test the fire spread probability adjustment. Confirm P_w yields higher probability when fire direction aligns with wind.

6. Experimental Results & Validation

6.1. Data Description

In this project, we investigate the spread of wildfire in 10 counties in the Los Angeles area, as shown in Figure 1 below. The recent wildfire in the Los Angeles area inspired us to use this location as our study region. The data used for this experiment include NDVI, NDDI, land cover type, slope, proximity to road, wind speed, wind direction, surface temperature, lightning density (synthetic), and wildfire history (synthetic). All the other datasets were obtained via Google Earth Engine except for lightning density and wildfire history. The NDVI was derived from Landsat 09 level 2 data product using spectral bands 5 (Near Infrared Band) and 4 (Red band). The NDVI data was normalized and classified into low ($NDVI \leq 0.2$), medium ($NDVI \leq 0.5$), and high-density ($NDVI > 0.5$) categories. We applied weights of -0.4, 0, and 0.3 from the literature to low, medium, and high-density categories. The NDDI was computed from the Normalized Difference Wet Index (NDWI) and NDVI. We clipped the computation result for the NDDI to ensure it falls between -1 and 1. The surface temperature was derived from Landsat 09 level 2 data product using the ST band 10. Also, the Euclidean distance approach was used to compute the distance to the road network in our study area. The road network was obtained from the U.S Census Bureau's Tiger program.

The land use dataset was obtained from Google's Dynamic World land use/land cover data, while wind speed and wind direction were obtained from the Open-Meteo API. Finally, the lightning density and historical wildfire were derived using a simple approach identified in the literature. The lightning density dataset was randomly generated using a scale (decay factor) of 0.3 from the exponential distribution. The historic wildfire dataset was randomly generated using an array of 0 and 1, with 1 indicating fire occurrence and 0 otherwise.

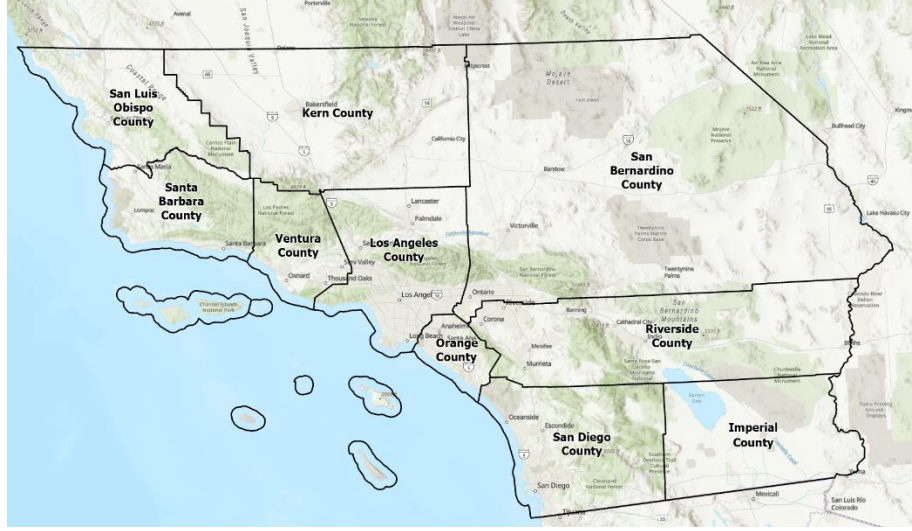


Figure 1: Map of our study area showing the 10 counties near Las Angeles, California that were included in our simulation.

6.2 Experimental Procedure

The first step in the experimentation procedure is the estimation of ignition probability. Before computing the ignition probability, we transformed all the raster data using the nearest neighbor resampling technique to ensure that the grid cells of all the rasters were the same. Next, we transformed all the datasets except land use using the min/max approach. We then assigned appropriate weight to land use depending on the flammability of the land use type. Higher weight indicates higher flammability, and zero indicates non-flammable. We used the beta parameters for each input data to estimate the ignition probability, as shown in the table below.

Input Data	Beta Parameter
Vegetation density (NDVI)	0.25
Terrian slope	0.15
Proximity to road	0.1
Land use	0.15
Historic wildfire	0.1
Surface Temperature	0.1
NDDI	0.2
Lightning density	0.2

Table 1: Beta parameter used to estimate ignition probability.

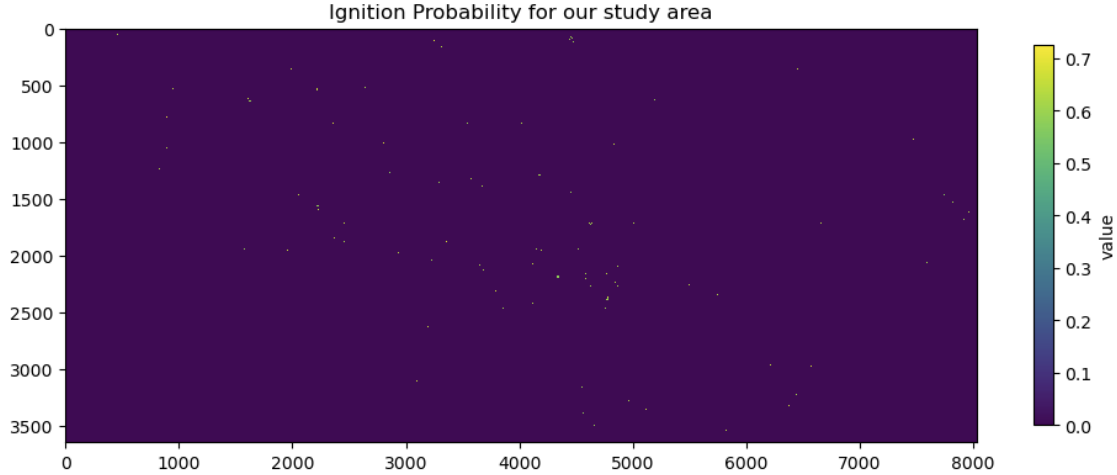


Figure 2: The ignition probability for our study area.

After computing the ignition probability, we selected five grid cells with the highest ignition probability as the ignition point. We initiated the start of the fire from these five grid cells. Once the fire starts, we use the wildfire probability spread equation to update the states of the grid cells. In computing the wildfire probability spread values for each of the eight neighbors of a given grid cell, we used land use, terrain slope, vegetation density, wind speed, and wind direction as the input dataset. Based on the literature, we assigned a weight to land use depending on the flammability of the land use type; a higher weight indicates highly flammable, while a negative or zero weight indicates non-flammable or non-burnable. Land use, such as water bodies, was treated as empty cells or cells with no fuel to propagate the fire. Next, we used 0.58 as the base probability of fire spread, 0.045 as the wind speed coefficient, 0.131 as the wind direction coefficient, and 0.078 as the terrain slope coefficient. These values were obtained from Alexandridis, et al. [20]. After we computed the probability of the wildfire spread, we compared it against a randomly generated value. If the spread probability of the grid cell is less than this value, we update its state to burning. Once a grid cell burns, we burn it out after each step and convert it to burned. Burned cells will remain in this state throughout the run of the simulation model. Finally, we experimented with waiting for three time steps before updating cells from burning to burned. The results of these experiments are presented in the experimental results section. The experimental results were analyzed using descriptive statistics and visualization.

6.3 Model Validation

6.3.1 Ignition Model Validation

Validating the wildfire ignition model ensures its predicted ignition probabilities align with observed fire starts. This section details the validation process using the Area Under the Receiver Operating Characteristic Curve (AUC) metric, which evaluates the model's ability to distinguish ignition-prone locations. The process involves obtaining ground-truth ignition data, comparing

predicted versus actual ignition locations, and conducting a sensitivity analysis to assess parameter robustness.

1. Obtaining Ground-Truth Ignition Data

Ground-truth data consists of historical wildfire ignition points, the potential source are:

- **California Department of Forestry and Fire Protection (Cal Fire):** The Fire and Resource Assessment Program (FRAP) database offers ignition point data, including coordinates and causes (e.g., lightning, human).
- **Satellite Products:** MODIS and VIIRS active fire detections provide ignition locations via thermal anomalies, filtered to exclude non-ignition hotspots.
- **National Interagency Fire Center (NIFC):** Supplies historical fire occurrence data with ignition coordinates.

2. Comparing Predicted vs. Actual Ignition Locations Using AUC

The ignition model outputs a 50x50 probability map, where each cell's value (0 to 1) represents the likelihood of ignition based on factors like vegetation density and slope. Validation compares this map to actual ignition locations using the AUC metric.

Actual ignition points are rasterized into a binary mask (M_{act})

$M_{act}(i, j) = 1$ if cell(i, j) is BURNED or BURNING

$M_{act}(i, j) = 0$ otherwise

The probability map P_{pred} and mask M_{act} are flattened into vectors.

The ROC curve plots the true positive rate (TPR) against the false positive rate (FPR) across thresholds:

$$TPR = \frac{True\ Positives}{True\ Positives + False\ Negatives}$$

$$FPR = \frac{False\ Positives}{False\ Positives + True\ Negatives}$$

True positive are ignition cells correctly predicted as high-probability; false positives are non-ignition cells incorrectly predicted. The AUC, ranging from 0 to 1, is the area under the ROC curve, where:

$AUC \cong 1$: Excellent discrimination

$AUC \cong 0.5$: Random guessing

$AUC < 0.5$: Worse than random

3. Sensitivity Analysis

Sensitivity analysis evaluates the model's robustness to input parameters (e.g., vegetation density, slope, lightning density). Each parameter is varied by $\pm 10\%$, or simplified inputs (e.g., constant lightning density) are tested against spatially varying alternatives. The probability map is recomputed for each scenario, and AUC is recalculated using the same ground-truth data.

6.3.2 Fire Spread Model Validation

Validating the wildfire spread model to ensure its predictions align with real-world fire behavior. This section describes the process of comparing the simulated burnt area to the actual fire perimeter, provided as a GeoJSON polygon, using the Intersection over Union (IoU) metric. IoU quantifies the spatial overlap between the predicted and observed burnt areas, offering a clear measure of model accuracy.

Validation Process

The validation involves aligning the simulated and actual fire data, computing IoU, and visualizing the results. Below are the key steps:

1. Preparing Simulated Burnt Area

The simulation outputs a grid of size 50×50 , where each cell represents a spatial unit ($\sim 193 \times 193$ meters). Cells marked as BURNED or BURNING indicate the predicted burnt area. This grid is converted into a binary mask (M_{sim}), defined as:

$$M_{sim}(i, j) = 1 \text{ if cell}(i, j) \text{ is BURNED or BURNING}$$
$$M_{sim}(i, j) = 0 \text{ otherwise}$$

2. Loading Actual Fire Perimeter

The actual fire perimeter is loaded from a GeoJSON file using the geopandas library. The file contains one or more polygons delineating the observed burnt area, typically derived from satellite imagery or fire agency records. The GeoJSON is assumed to use the WGS84 coordinate system (EPSG:4326). If its coordinate reference system differs, it is reprojected to WGS84 for consistency with the simulation grid.

3. Rasterizing the Actual Perimeter

To enable comparison, the GeoJSON polygon is rasterized onto the same 50×50 grid as the simulation. This creates a binary mask (M_{act}) where cells inside the polygon are assigned 1, and those outside are 0. Rasterization uses the grid's spatial bounds, mapping latitude and longitude to pixel coordinates via a transform matrix. The `rasterio.features.rasterize` function ensures accurate projection of the polygon onto the grid. For multiple polygons, all are combined into a single mask.

4. Calculating Intersection over Union (IoU)

The IoU metric measures the spatial agreement between the simulated (M_{sim}) and actual (M_{act}) burnt areas. It is defined as:

$$IoU = \frac{|M_{sim} \cap M_{act}|}{|M_{sim} \cup M_{act}|}$$

Where:

Intersection ($|M_{sim} \cap M_{act}|$): The number of cells where both masks are 1, calculated as:

$$|M_{sim} \cap M_{act}| = \sum_{i=1}^n \sum_{j=1}^n (M_{sim}(i, j) \cdot M_{act}(i, j))$$

Union ($|M_{sim} \cup M_{act}|$): The number of cells where at least one mask is 1, calculated as:

$$|M_{sim} \cup M_{act}| = \sum_{i=1}^n \sum_{j=1}^n (M_{sim}(i, j) + M_{act}(i, j) - M_{sim}(i, j) \cdot M_{act}(i, j))$$

IoU ranges from 0 (no overlap) to 1 (perfect overlap), with higher values indicating better model accuracy.

5. Visualizing Results

The simulated and actual burnt areas are visualized by overlaying their masks. The simulated mask is plotted in red (semi-transparent) and the actual mask in blue (semi-transparent), using the grid's latitude and longitude extents. Overlapping regions appear purple, highlighting agreement. The IoU value is displayed in the plot title, aiding qualitative assessment.

Interpretation

An IoU close to 1 indicates strong spatial alignment, suggesting the model accurately predicts the fire's extent. Values around 0.5–0.7 suggest moderate agreement, with some overlap but potential errors in shape or location. Low IoU (<0.3) indicates significant discrepancies, possibly due to model limitations or data misalignment. The visualization complements IoU by revealing patterns in errors, such as overprediction or underprediction in specific directions.

Assumptions

- **Grid Resolution:** The 50 x 50 grid assumes uniform cell areas, sufficient for the ~10 km x 10 km extent but potentially coarse for complex perimeters.
- **Data Alignment:** The GeoJSON polygon is assumed to cover the simulation area, ensuring meaningful comparison.
- **Binary Representation:** Both burnt areas are treated as binary, simplifying partial burns or intensity variations.

To evaluate our wildfire simulation model, we conducted experiments focused on simulating fire spread from known ignition points, rather than attempting to model both ignition and spread

simultaneously—since real-world fire reports typically contain only one ignition point or none at all, making full comparison difficult. We selected two wildfire events from 2023 for this test: the Rabbit Fire and the Rinconada Fire, which had documented ignition coordinates at (33.89148, -117.02139) and (35.0866, -120.01923), respectively. Starting from these ignition points, we ran the simulation for a duration of 48 hours and then calculated the Intersection over Union (IoU) between the simulated burned area and the actual fire perimeter. The Rabbit Fire yielded an IoU of 0.5285, indicating moderate spatial overlap, while the Rinconada Fire had a much lower IoU of 0.0516.

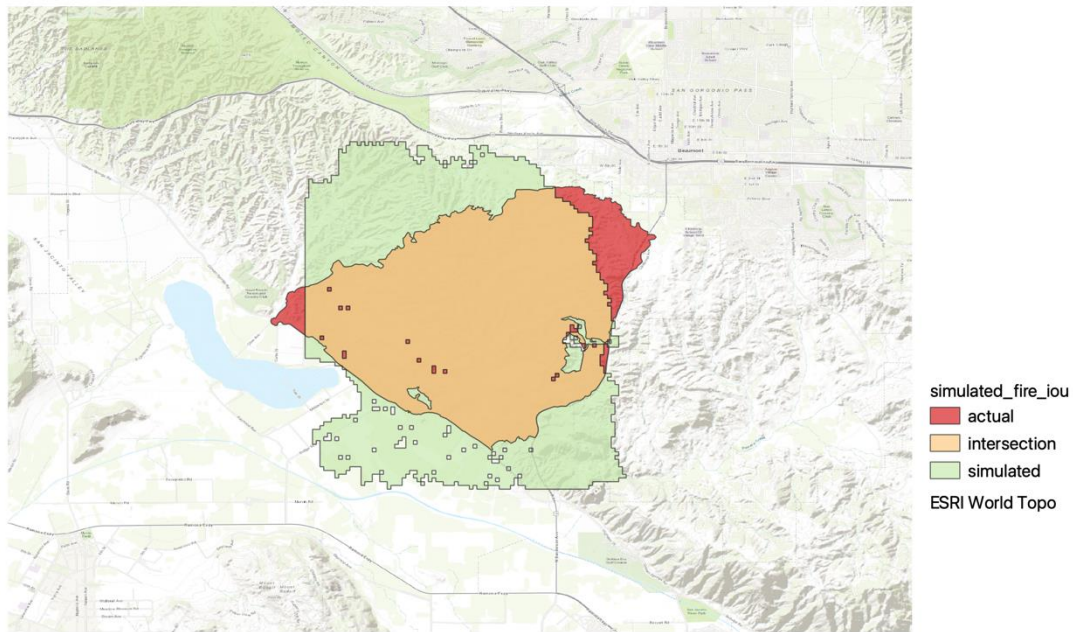


Figure 3: Rabbit Fire Simulation vs Actual.

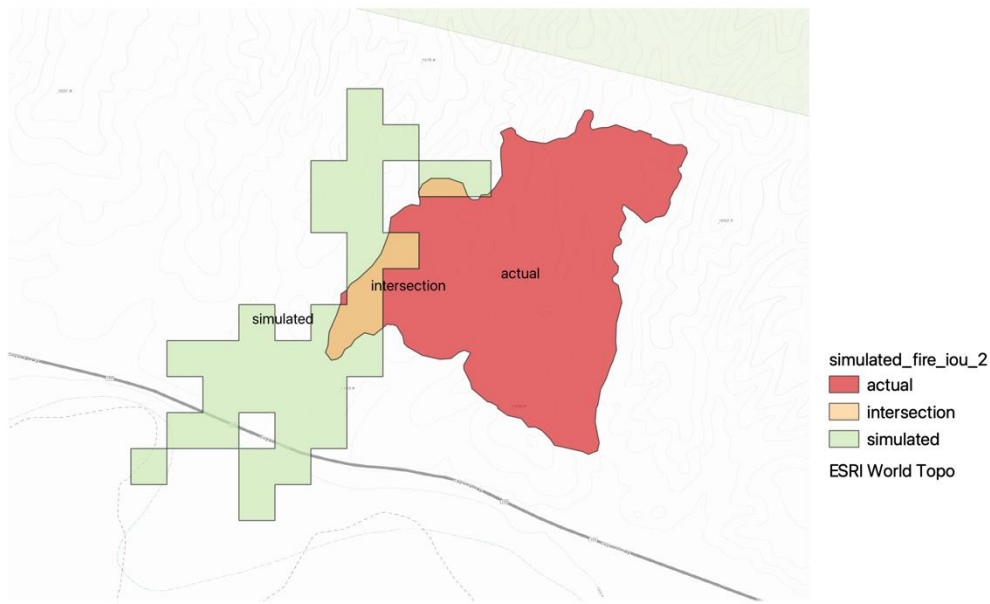


Figure 4: Rinconada Fire Simulation vs Actual.

Several factors may explain this difference in performance. The higher IoU for the Rabbit Fire suggests that the environmental conditions and terrain in that region may have been better captured by the model inputs (e.g., wind, vegetation, slope). Conversely, the low IoU for the Rinconada Fire could be due to inaccuracies in environmental data, local factors not reflected in the model (such as suppression efforts or microclimate effects), or the model's inability to fully capture the complexity of fire behavior in that area.

6.4 Experimental Results

6.4.1 Visualization of Simulation Results

To explore how burning and burned areas develop, we tested our model for $t = 0, \dots, 500$ and created a new land use dataset for time steps 30, 100, 300, and 500. In the first simulation run, burning cells were converted to burned for each step. Figures 5 to 8 present the results of the first simulation run. In the second simulation run, we delayed the conversion from burning to burned until the 4th step. Figures 9 to 12 present the results of the second simulation run. The figures indicate that the burned area grows exponentially in both simulation runs over time. The burning areas increase gradually and form a perimeter around the burned region. Additionally, the results from both simulation runs suggest that land uses such as built-up areas, crops/agriculture, and bare land are less flammable and act as fire breaks compared to trees or grassland. Consequently, the fire spread quickly when the land type was trees, followed by grassland. Water bodies serve as fire breaks, as illustrated in the figures below. Furthermore, the simulation results show that smaller land areas burn out quickly. From the figures below, one of the ignition points was in Santa Catalina Island, which burned out after 200 time steps since the predominant land use type consists of trees and grassland. The spread of fire from the island to the mainland is impossible

due to the ocean. Finally, comparing the results of the first and second simulation runs shows that wildfire spread faster in the first simulation run than in the second.

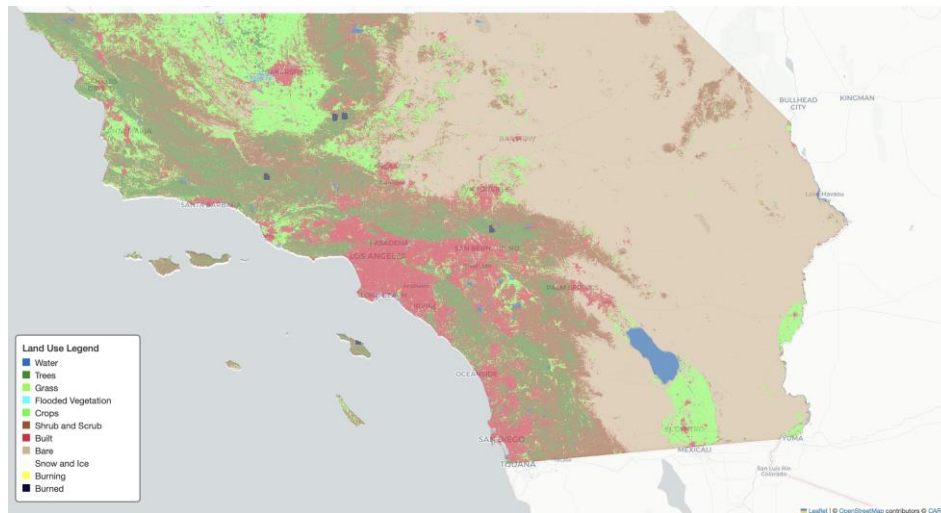


Figure 5: Visualization of 1st simulation result after initiating ignition from 5 different locations for 30 time steps.

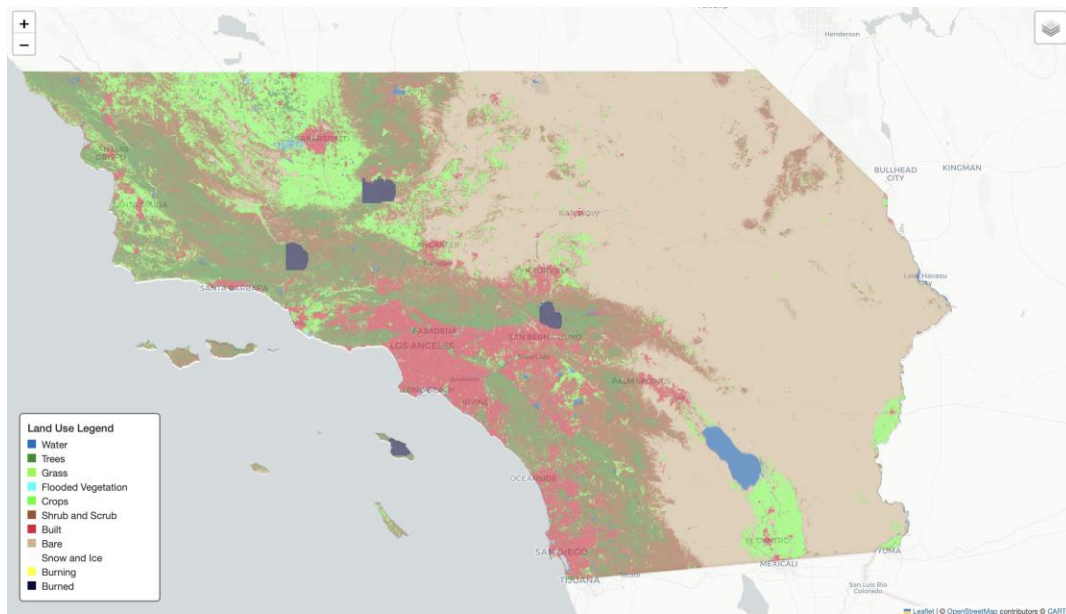
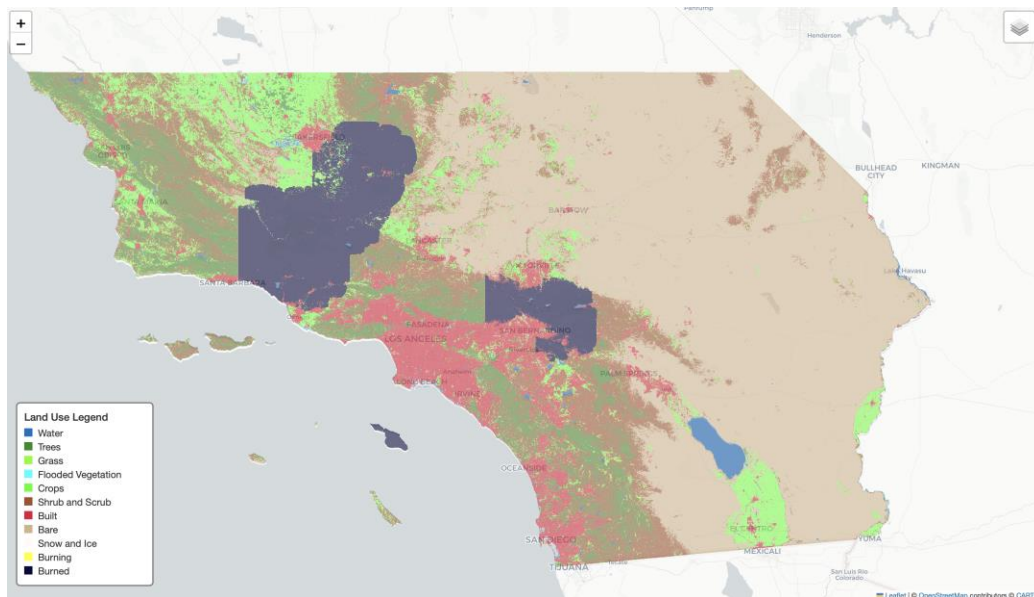
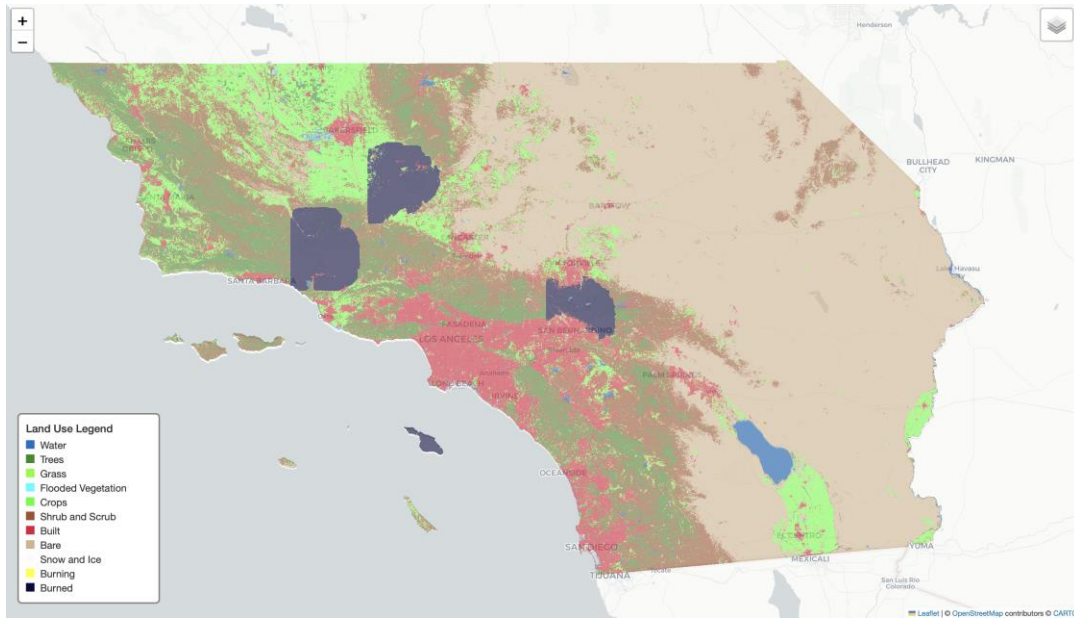


Figure 6: Visualization of 1st simulation result after initiating ignition from 5 different locations for 100 time steps.



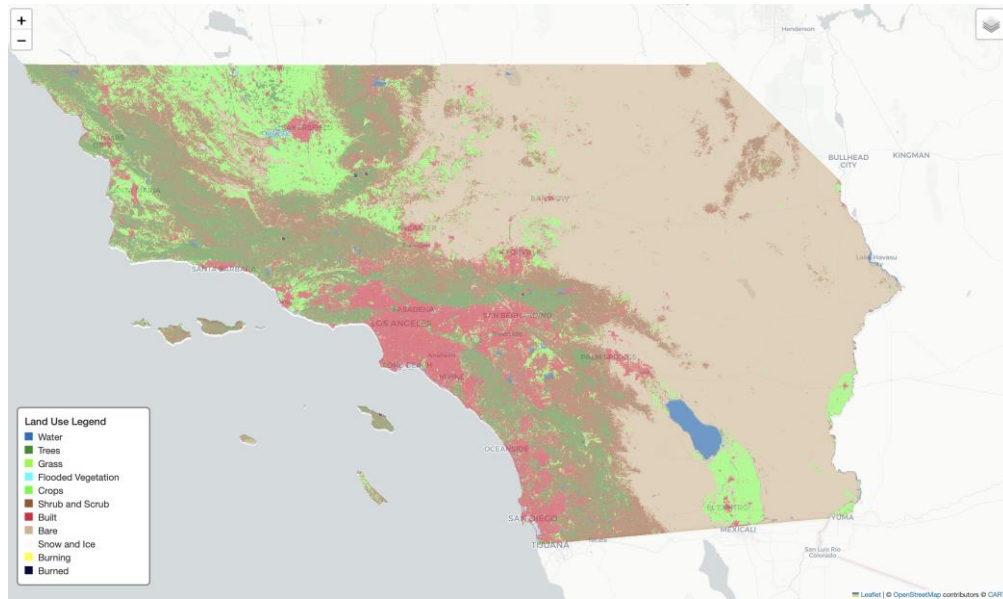


Figure 9: Visualization of 2nd simulation result after initiating ignition from 5 different locations for 30 time steps.

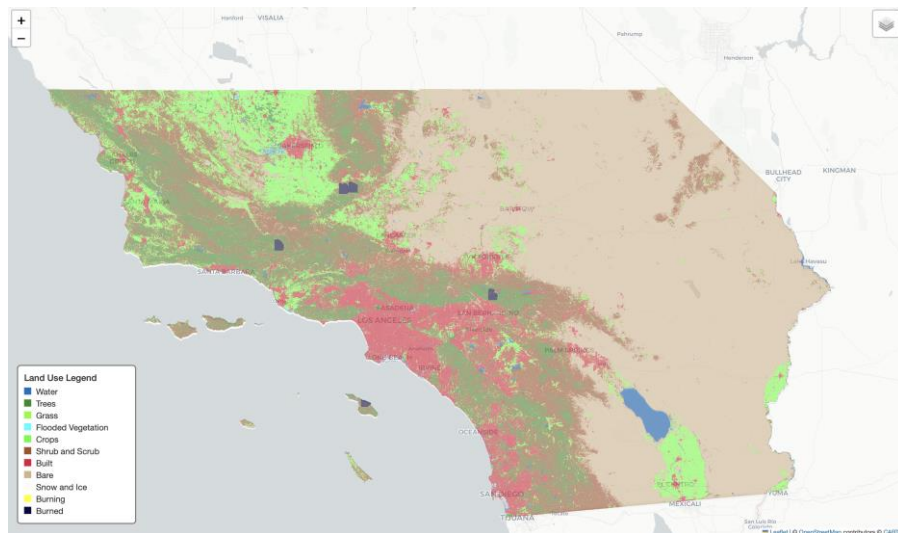


Figure 10: Visualization of 2nd simulation result after initiating ignition from 5 different locations for 100 time steps.

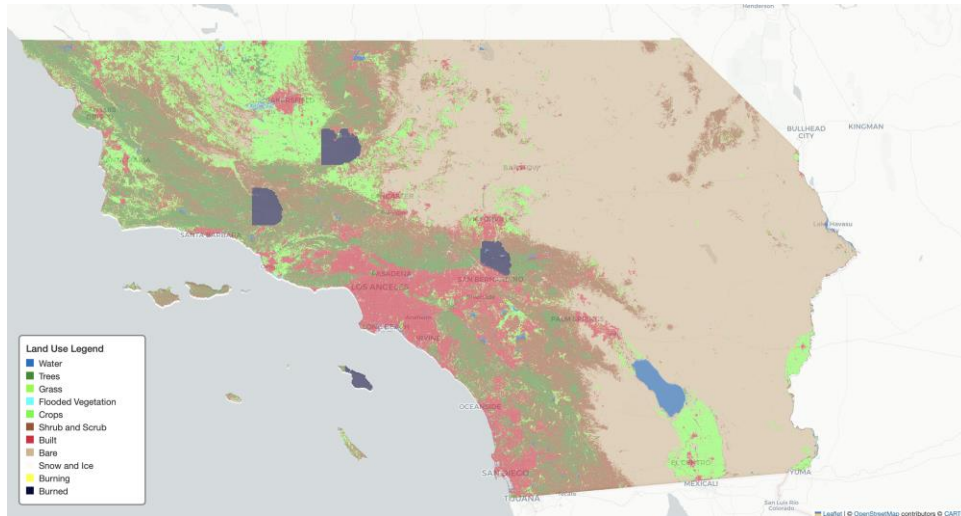


Figure 11: Visualization of 2nd simulation result after initiating ignition from 5 different locations for 300 time steps.

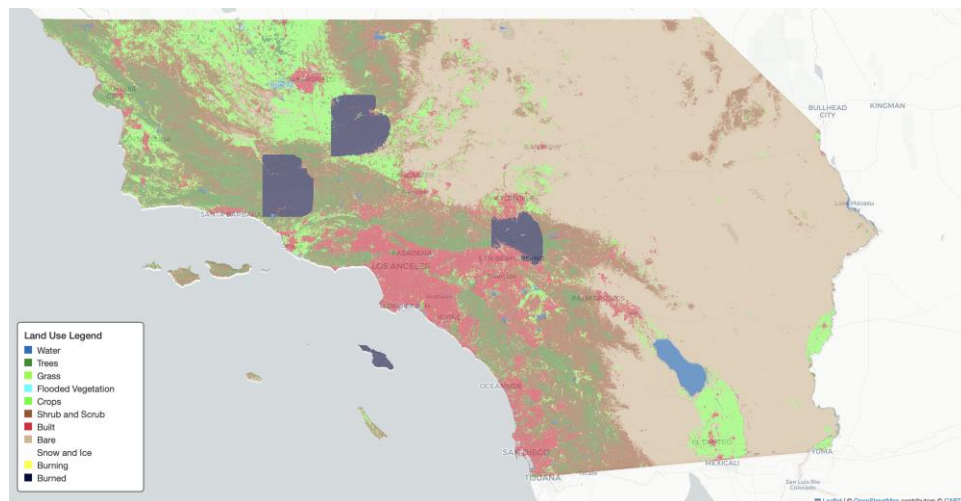


Figure 12: Visualization of 2nd simulation result after initiating ignition from 5 different locations for 500 time steps.

Figure 13 below shows that at every time step, the burning vs. burned area for the first simulation run is larger than that of the second. This further indicates that the wildfire spreads gradually in the second simulation than in the first.

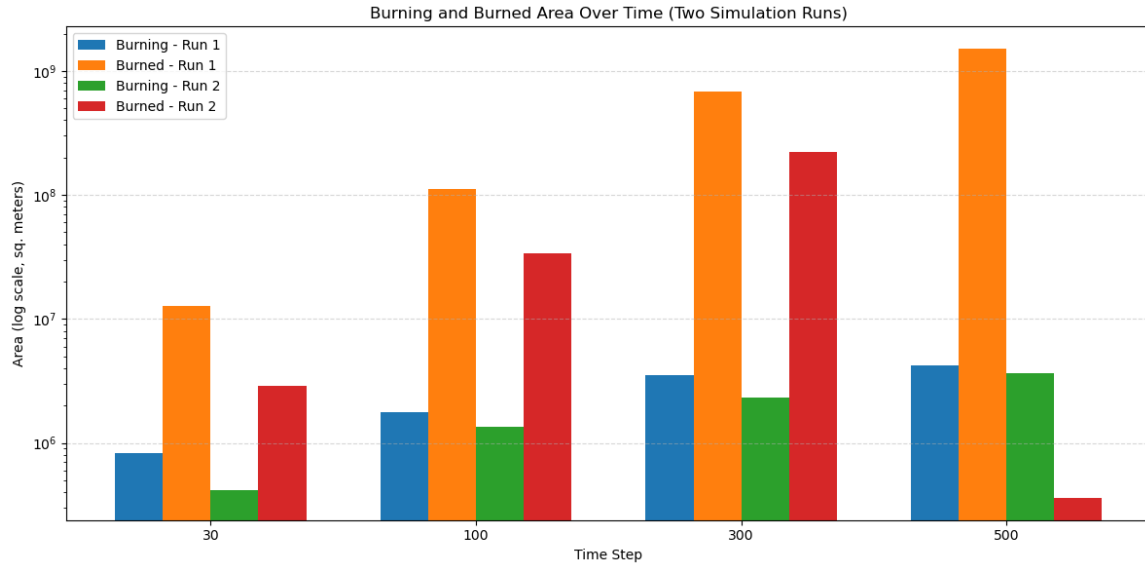


Figure 13: Burning and Burned Area Over Time (Two Simulation Runs).

6.4.2 Burning vs Burned Cell Count Over time

To compare the number of burning and burned cells, we tested our model for $t=0, \dots, 20$ and recorded the number of burning and burned cells at each step. We performed two simulation runs: one in which burning cells are converted to burned cells at each step, and another in which we impose a 3-step delay and update burning cells to burned cells on the fourth step. Figure 14 shows that for the first simulation run, the number of burned cells is higher than that of burning cells after the third step and grows exponentially. Figure 15 shows that the delay causes the number of burned cells to accumulate gradually for the second simulation run. Still, it increases more than the number of burning cells after the eleventh step.

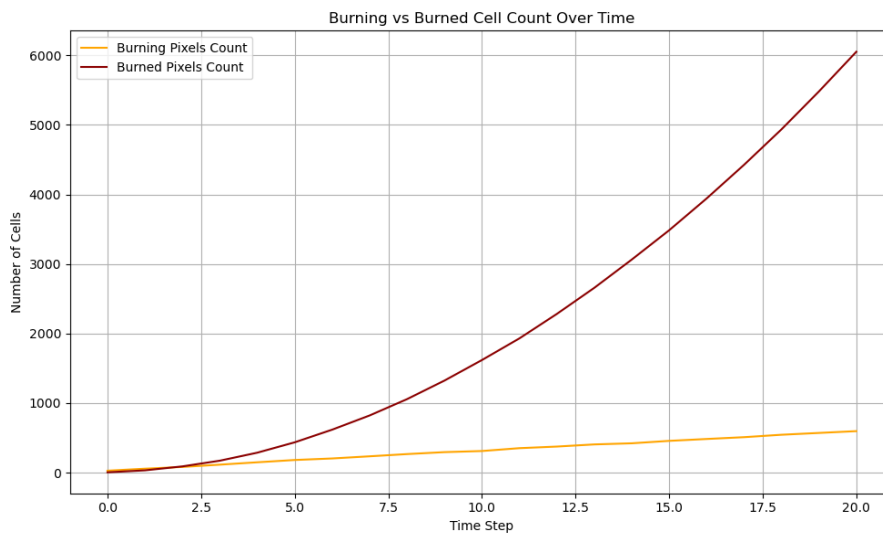


Figure 14: Burning vs. burned cell count over time in the simulation. Burning cells are converted to burned after each time step, resulting in a steadily increasing number of burned cells and a slower rise in active burning cells.



Figure 15: Burning vs. burned cell count over time with a 3-step delay before conversion. The delay causes burning cell counts to fluctuate while burned cells accumulate more gradually in discrete jumps.

6.4.3 Burned Area by Fuel Type

The figure below shows burned area by fuel type. We tested our model for $t = 0, \dots, 50$ and calculated the burned area by fuel type. We only show the result of the first simulation run, where we converted burning cells to burned after each time step. Since both simulations produce similar results, we did not show the result of the second simulation run, in which we delayed three steps before updating burning cells to burned cells. The figure shows that the burned area for forest/trees is the largest compared to all the other fuel types, followed by shrub grasslands.

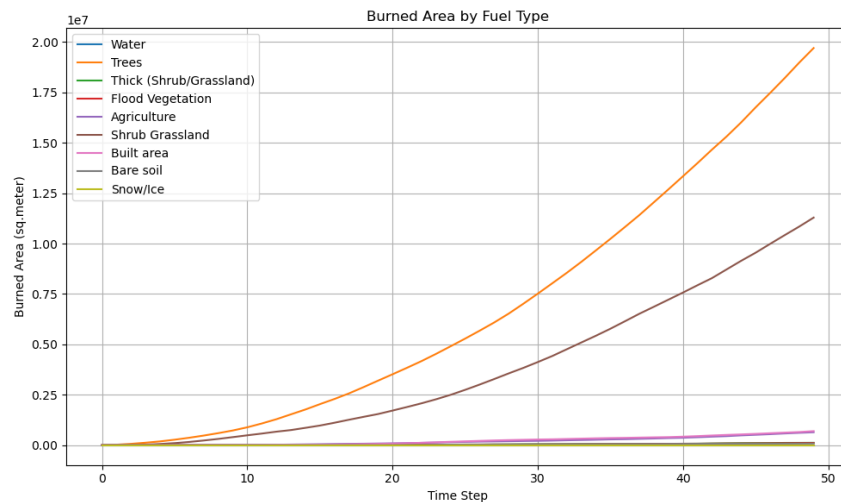


Figure 16: Burned area over time by fuel type. Forested areas and shrub grasslands exhibit the largest burned area, indicating their high flammability and abundance in the landscape compared to other land cover types.

6.4.4 Average Slope of Burned Pixels

To explore the effect of slope on fire propagation, we tested our model for $t = 0, \dots, 20$ and computed the average slope of the burned grid cell. Figure 17 indicates that fire starts from a steeper terrain and spreads gradually down the steep slope where forest vegetation is likely available.

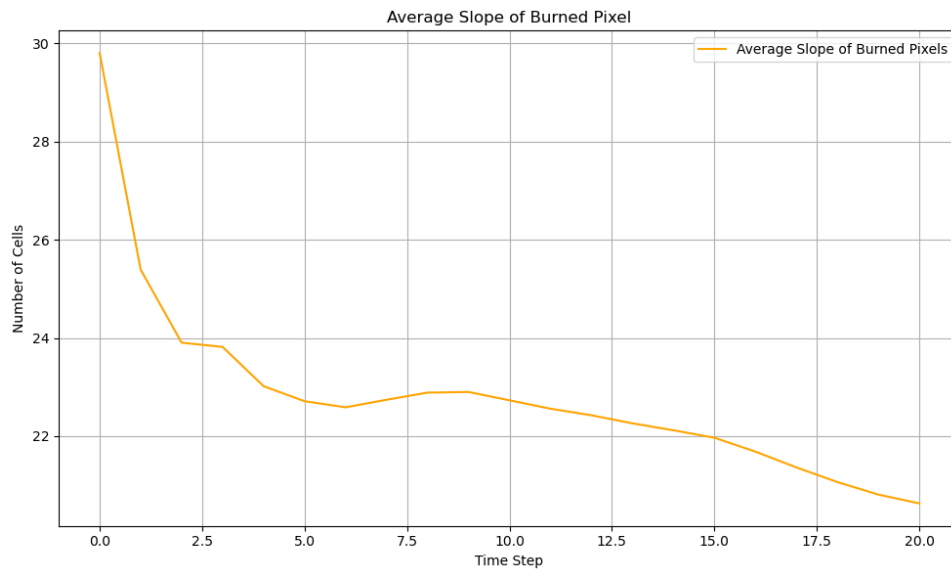


Figure 17: Average slope of burned pixels over time in the California case study. The fire initially spreads through steeper terrain, but as it expands, it progresses into flatter regions, resulting in a gradual decline in average slope.

7. Discussion

Our cellular automata-based simulation provided several key insights into wildfire behavior. By incorporating spatially explicit environmental variables—vegetation type, vegetation density, terrain slope, and wind speed/direction—we observed how each factor shapes the emergent patterns of fire spread. For example, denser forest cells with high NDVI values dramatically accelerated front-line propagation, particularly when aligned with prevailing winds, while areas classified as shrubland or agriculture acted as partial barriers. The probabilistic ignition map likewise highlighted regions of elevated fire risk tied to historical fire occurrences and anthropogenic influences. Overall, the exercise affirmed that wildfire dynamics cannot be captured by a single parameter. In fact, it's the nonlinear interplay among multiple unpredictable environmental drivers that dictates the speed, direction, and irregular shape of wildfire spread.

To build upon this work, future efforts could pursue three primary directions. First, integrating higher-resolution input data — such as sub-10 m seasonal NDVI rasters with dynamic

thresholds, detailed digital elevation models, moisture indices, and refined land-cover/land-use layers — would improve the spatial accuracy of spread predictions. Second, coupling the simulator to live data streams (e.g., near-real-time weather APIs, satellite-derived moisture indices) would support adaptive forecasts of fire behavior as conditions evolve. Finally, developing an interactive, web-based dashboard would enable stakeholders to visualize simulation outputs, adjust parameter settings, and run “what-if” scenarios to evaluate mitigation or evacuation strategies. Together, these extensions would enhance both the accuracy and practical applicability of the CA framework for wildfire modeling.

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Division of Labor:

The project was carried out through a clear division of labor. Thanawit and Gabriel led the development and refinement of the wildfire ignition model, with Thanawit also obtaining and integrating dynamic wind speed and direction data from an API to assess its influence on fire spread. Katherine was responsible for incorporating the effects of different vegetation types into the simulation framework, while Kshitij focused on modeling how varying vegetation density alters the propagation of wildfire across the grid. Gabriel further ensured that the simulation outputs could be accurately overlaid onto real geographic areas for visualization and analysis. Finally, the entire team collaborated to validate the integrated model against relevant datasets, confirming that each component behaved as expected within the probabilistic cellular automata framework.